MEMORANDUM

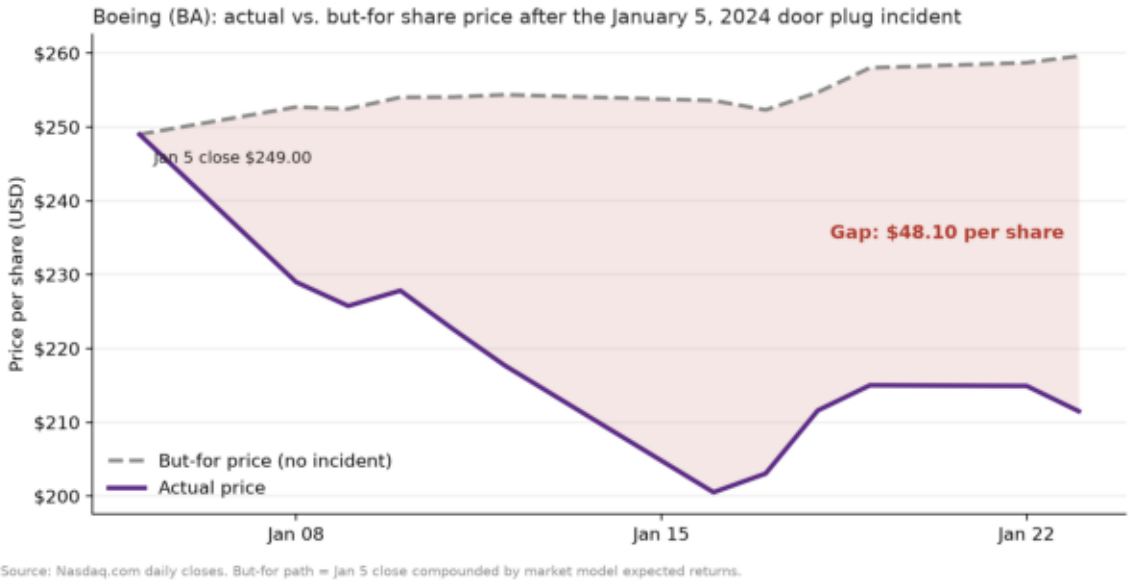
From: Erica Mathias | Re: Boeing stock price reaction to the Alaska Airlines Flight 1282 door plug incident

QUESTION

How much shareholder value did Boeing lose because of the January 5, 2024 Alaska Airlines door plug incident, separate from normal market movement?

ANSWER

On Monday, January 8, 2024, the first trading day after the incident, Boeing stock returned -8.03% while the market model predicted 1.48%. The difference, an abnormal return of -9.52%, is statistically significant (t = -6.13, p < 0.001). Applied to Boeing's market value, this is a change of -$14.46 billion in a single day. By January 23, 2024, the actual price was $211.50 against a but-for price of $259.60, a gap of $48.10 per share. The day 0 result holds across 6 alternative model setups (abnormal return -9.76% to -9.38%).



Source: Nasdaq.com daily closes. But-for path = Jan 5 close compounded by market model expected returns.

CUMULATIVE ABNORMAL RETURN BY EVENT WINDOW

Window	Dates	Abnormal return	t-stat	p-value	Change in market value
[0]	Jan 8	-9.52%	-6.13	< 0.001	-$14.46 billion
[0, +1]	Jan 8 to Jan 9	-10.84%	-4.94	< 0.001	-$16.30 billion
[0, +5]	Jan 8 to Jan 16	-22.75%	-5.99	< 0.001	-$32.33 billion
[0, +10]	Jan 8 to Jan 23	-19.67%	-3.82	< 0.001	-$28.68 billion

Day 0 is the headline result. Longer windows are shown as sensitivity checks and mix in later news.

METHOD

I estimated how Boeing normally moves with the overall market by regressing its daily return on the S&P 500 ETF (SPY) over 250 trading days (November 23, 2022 to November 21, 2023), ending 30 trading days before the event so pre-event news does not distort the model (beta = 1.0003, R-squared = 0.257). That model gives a but-for return for each day after the incident: what Boeing would have returned given the market's move if the incident had not happened. The abnormal return is the actual return minus the but-for return, tested against the model's normal daily noise (1.55%). The dollar figure multiplies each day's abnormal return by the prior close and 610,135,205 shares outstanding (FY2023 Form 10-K).

LIMITATIONS

- The FAA grounded 171 MAX 9 jets on Saturday, January 6, so day 0 reflects the incident and the grounding together. Later news (the FAA investigation and production oversight in the following days) overlaps the longer windows, so multi-day results mix several events.
- A single-factor market model. A fuller analysis could add an aerospace or industrials index.
- SPY closing prices are not dividend-adjusted. The effect is negligible over these windows.
- This is an educational analysis of public data, not a legal damages opinion. Shareholder damages in a real case depend on trading patterns, class periods, and legal theory.